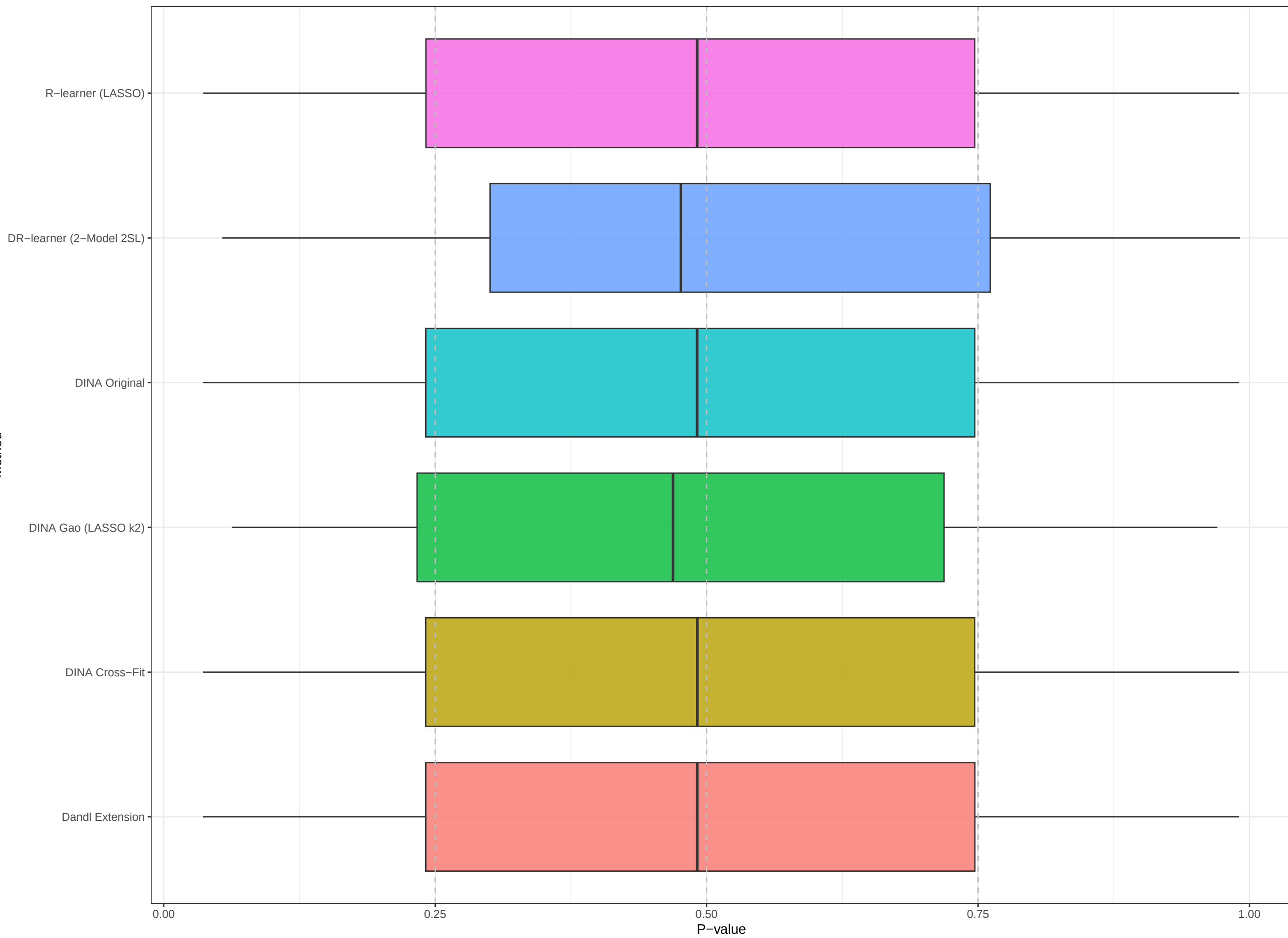
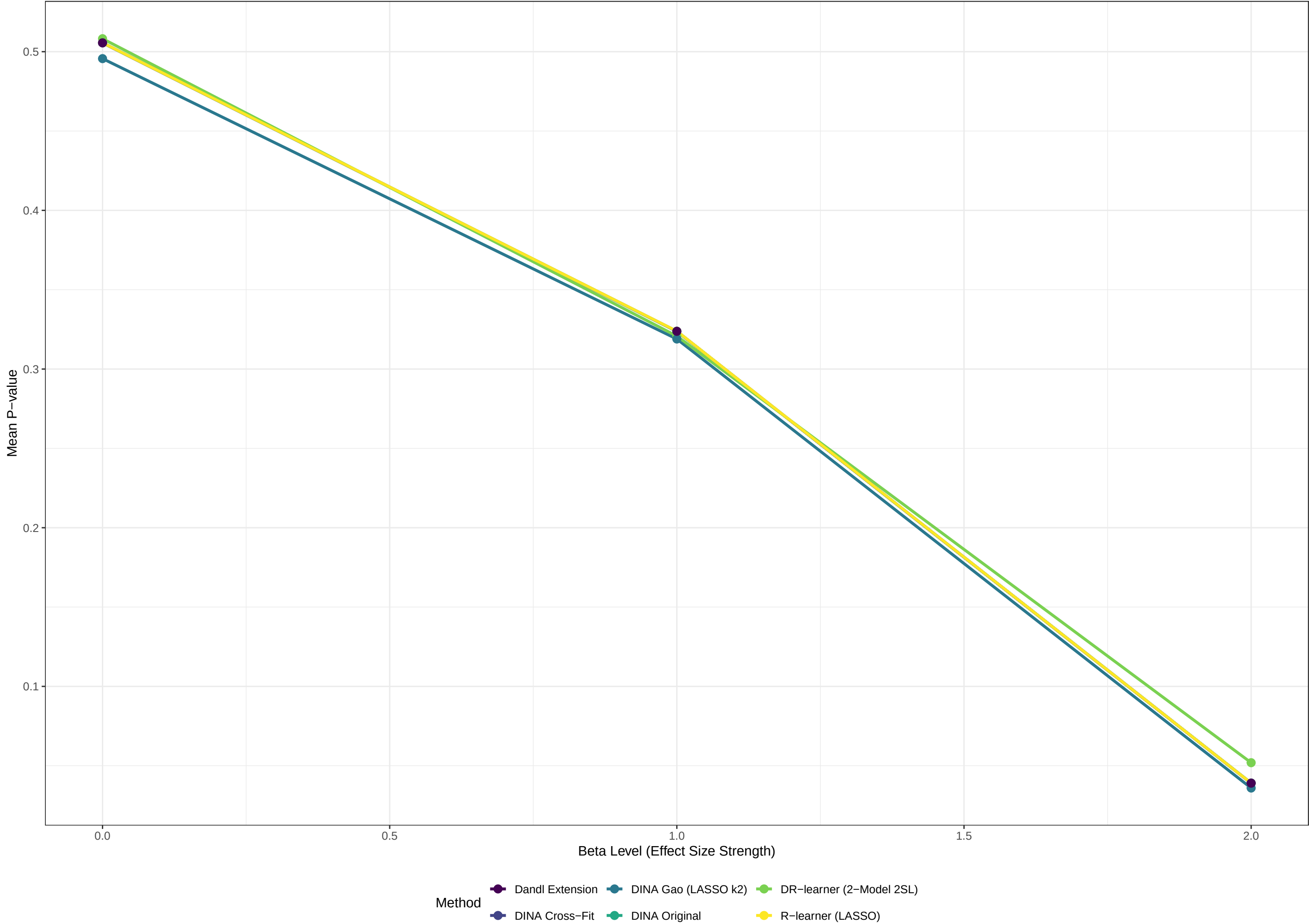


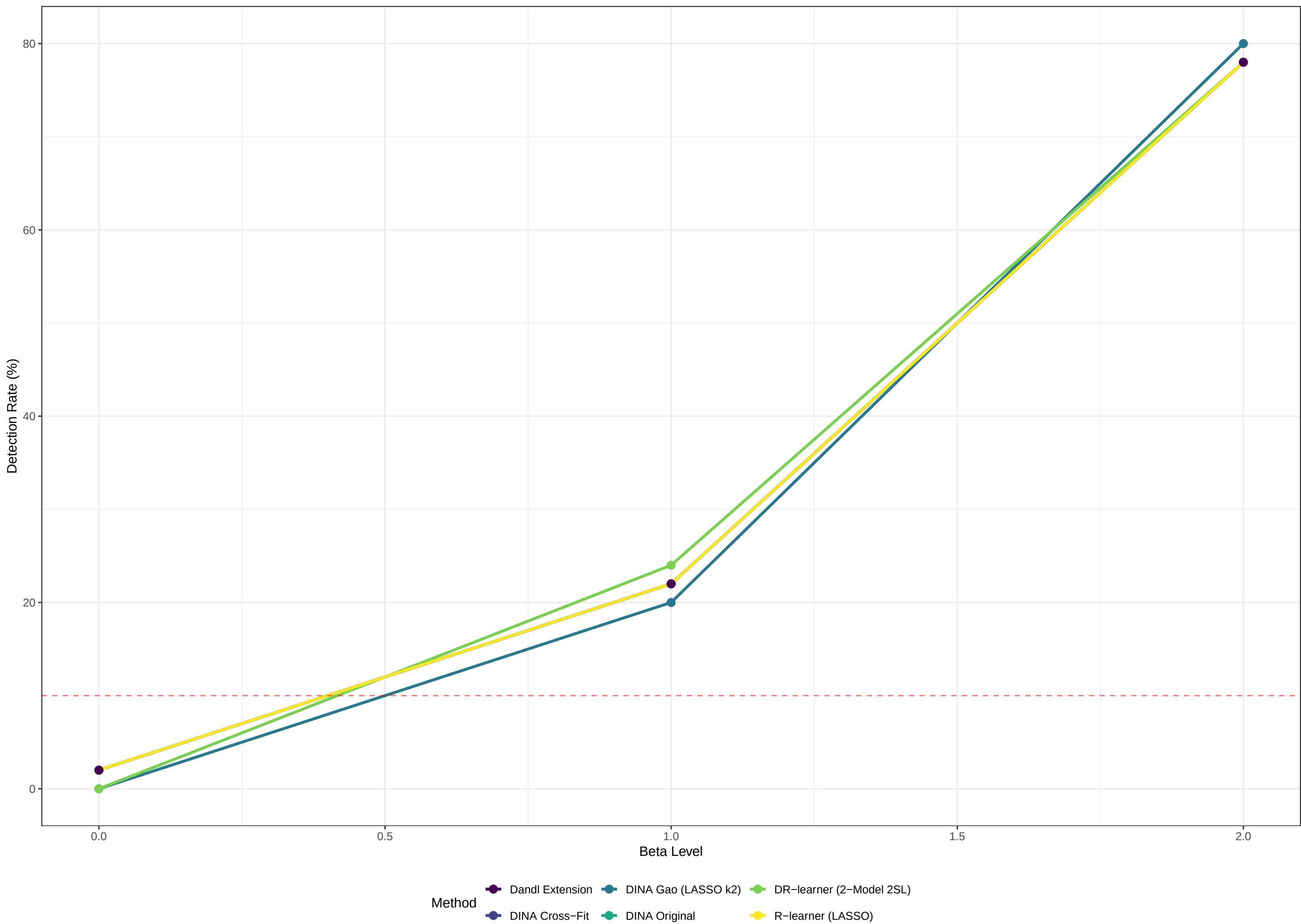
Objective 1(i): P-value Under Null (Scenario 2, $\beta=0$)
P-values should be $\sim \text{Uniform}[0,1]$ when no heterogeneity exists



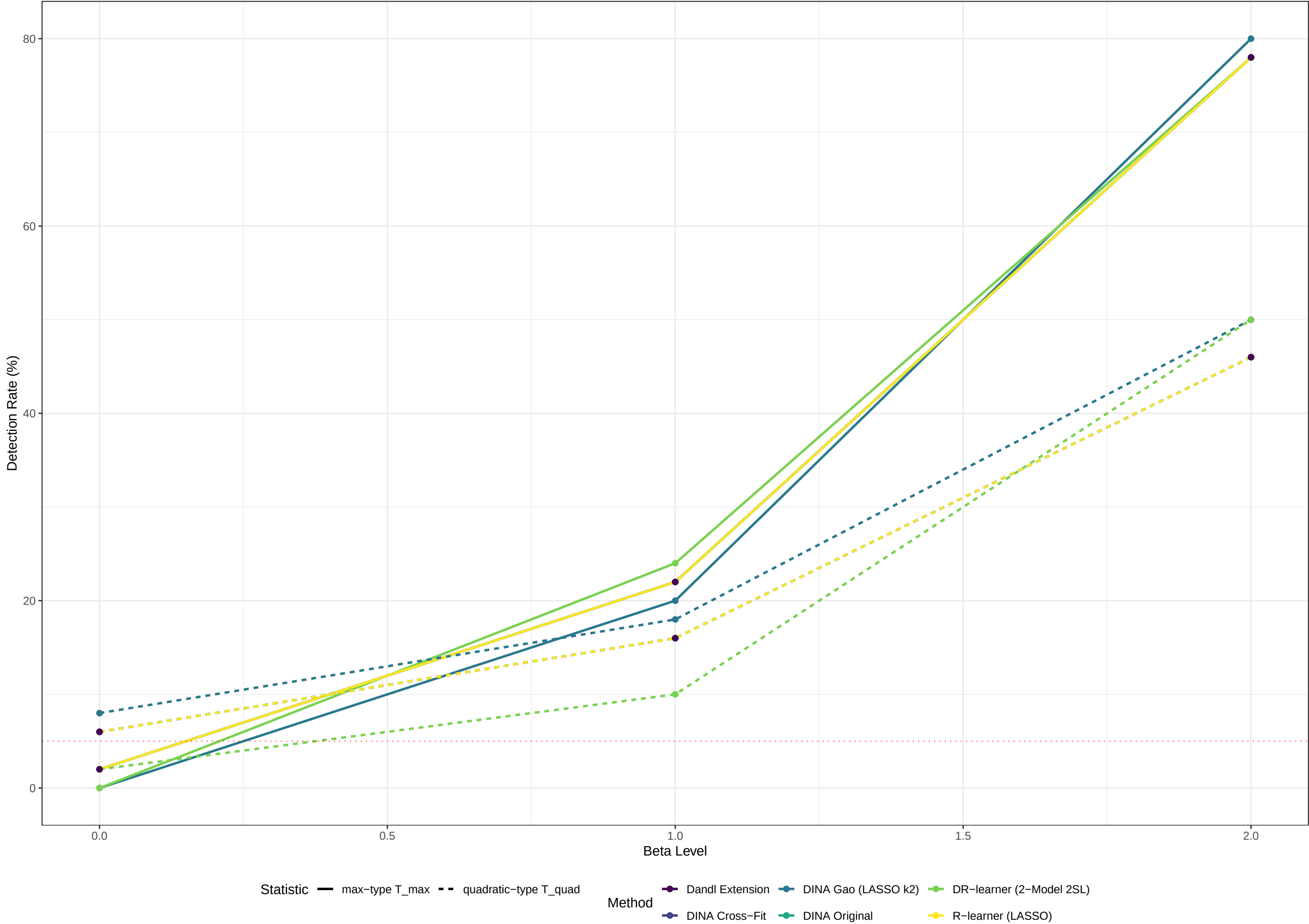
Objective 1(ii): Power Analysis (Scenario 2)
Lower mean p-value indicates better power to detect heterogeneity



Objective 1: Detection Rate (Scenario 2)
% of sims detecting heterogeneity (p < 0.05). Red line = nominal ..

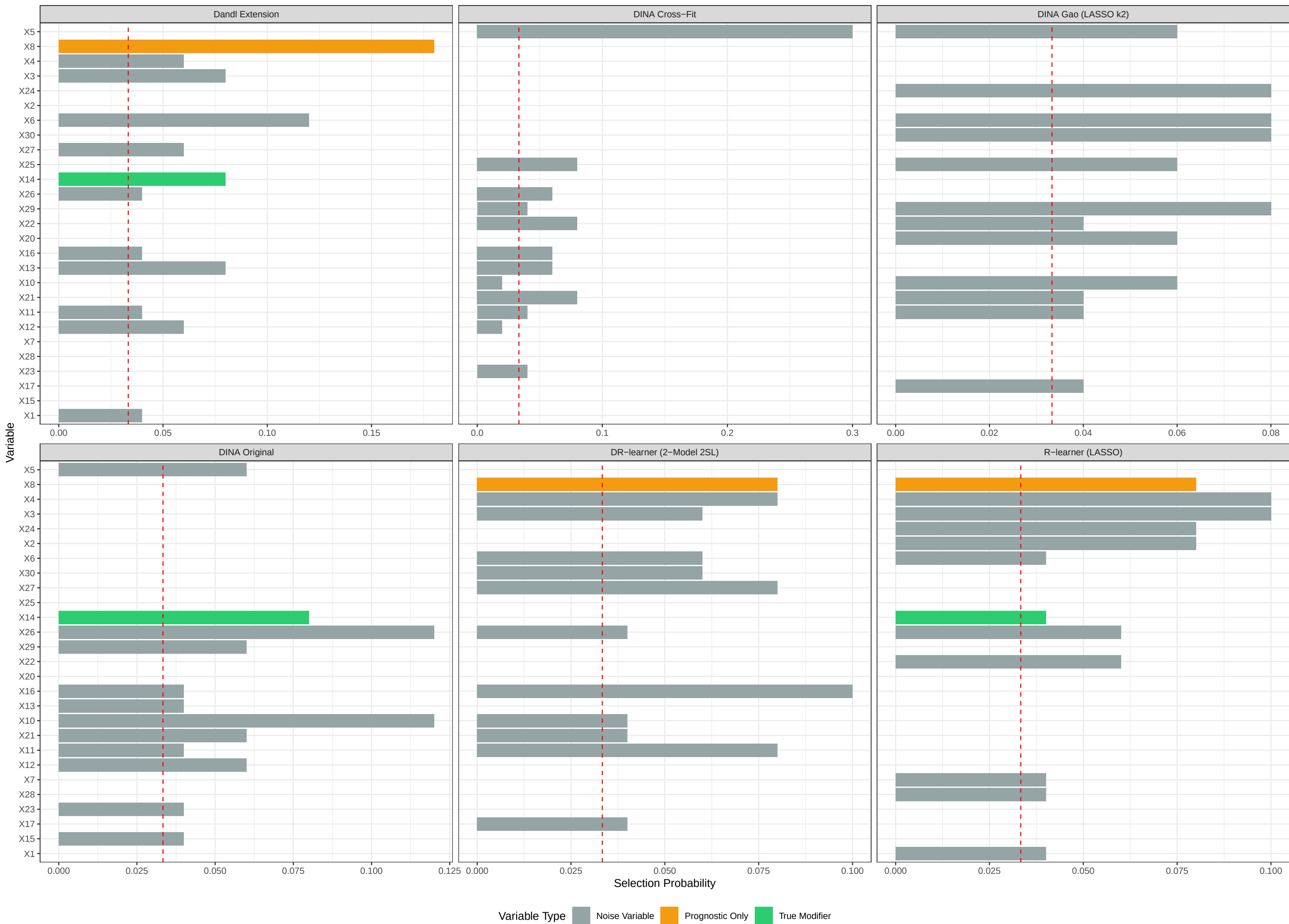


Objective 1: Max- vs Quadratic-type Power (Scenario 2)
Permutation test on each method's pseudo-outcome. Red dotted = nominal $\alpha = 5\%$.



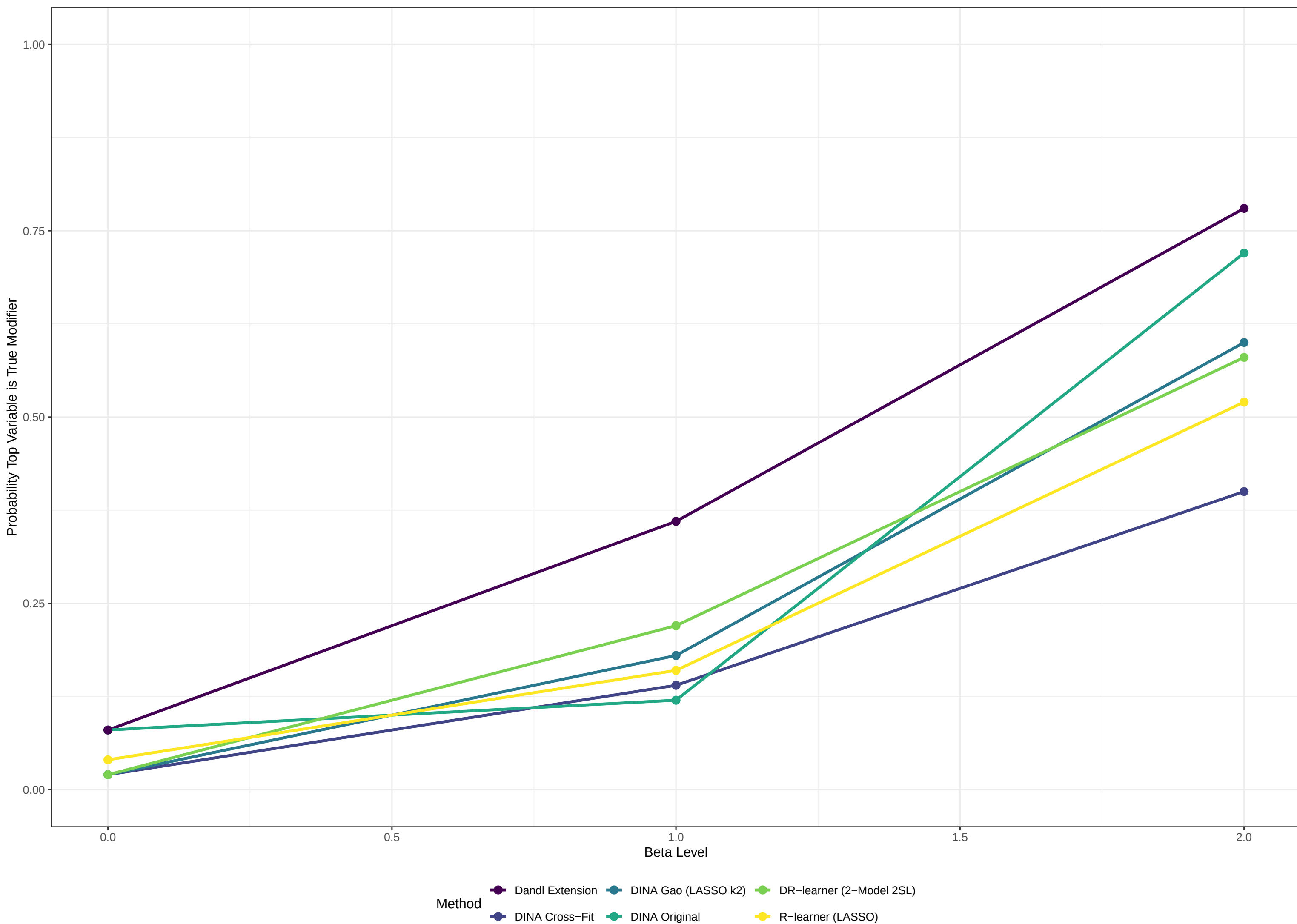
Objective 2(i): Top Modifier Under Null (Scenario 2, .=0)

Unbiased methods select all vars equally (~3.3% for 30 vars, red line)



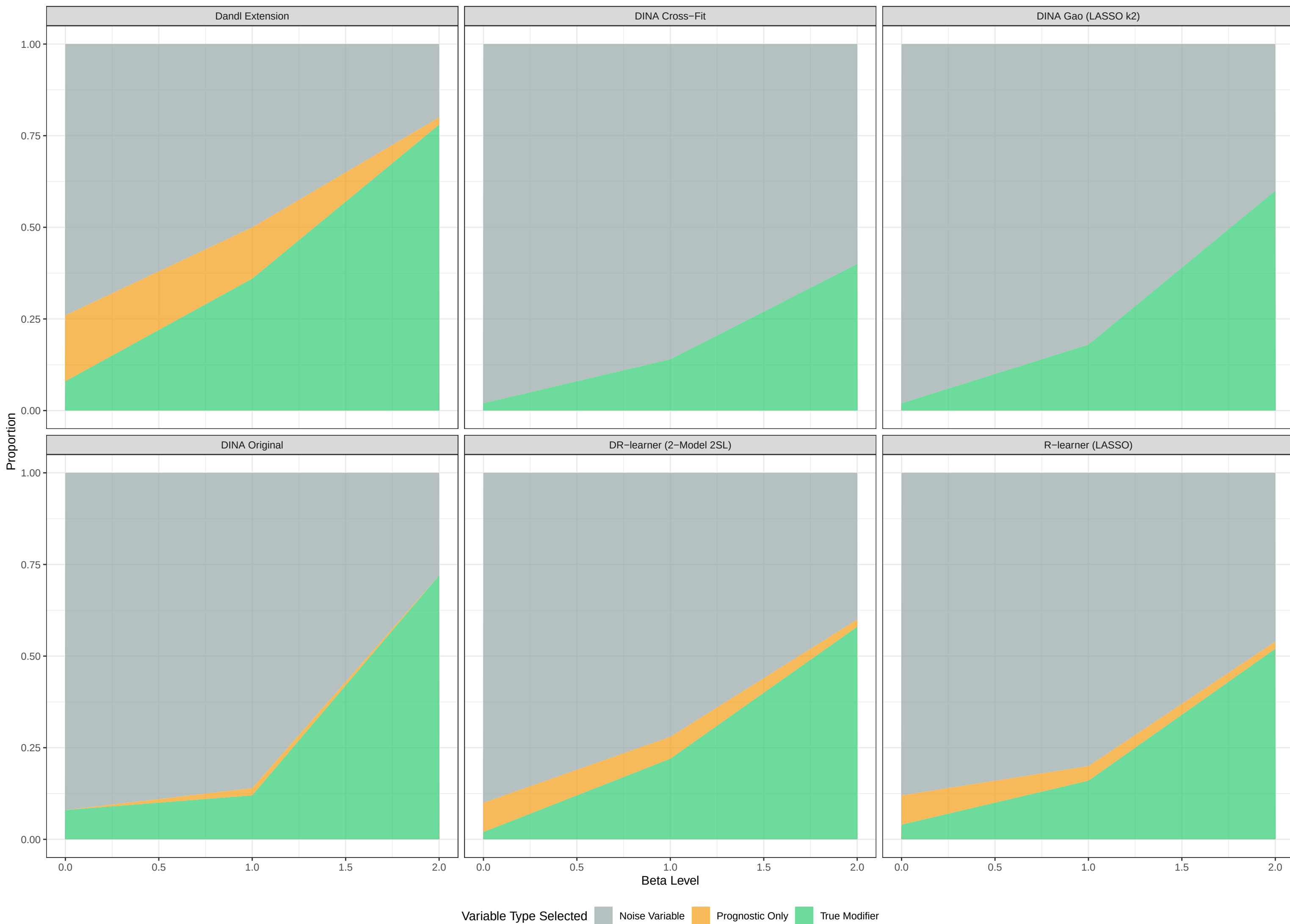
Objective 2(ii): Prob Top Variable is Truly Predictive (Scenario 2)

True modifiers: X14



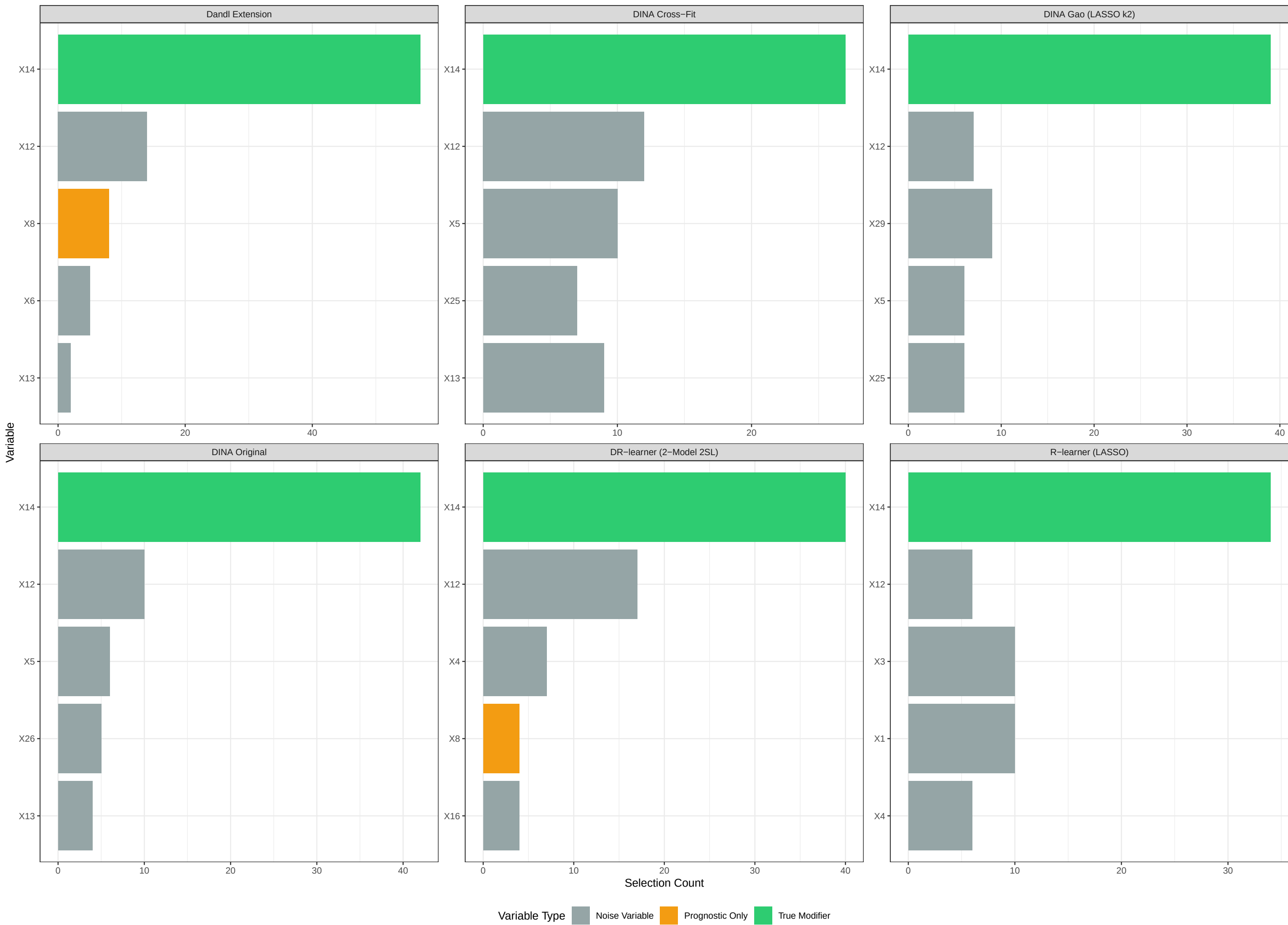
Objective 2(ii): Variable Type Selection Breakdown (Scenario 2)

Stacked area: what type of variable was selected as top modifier



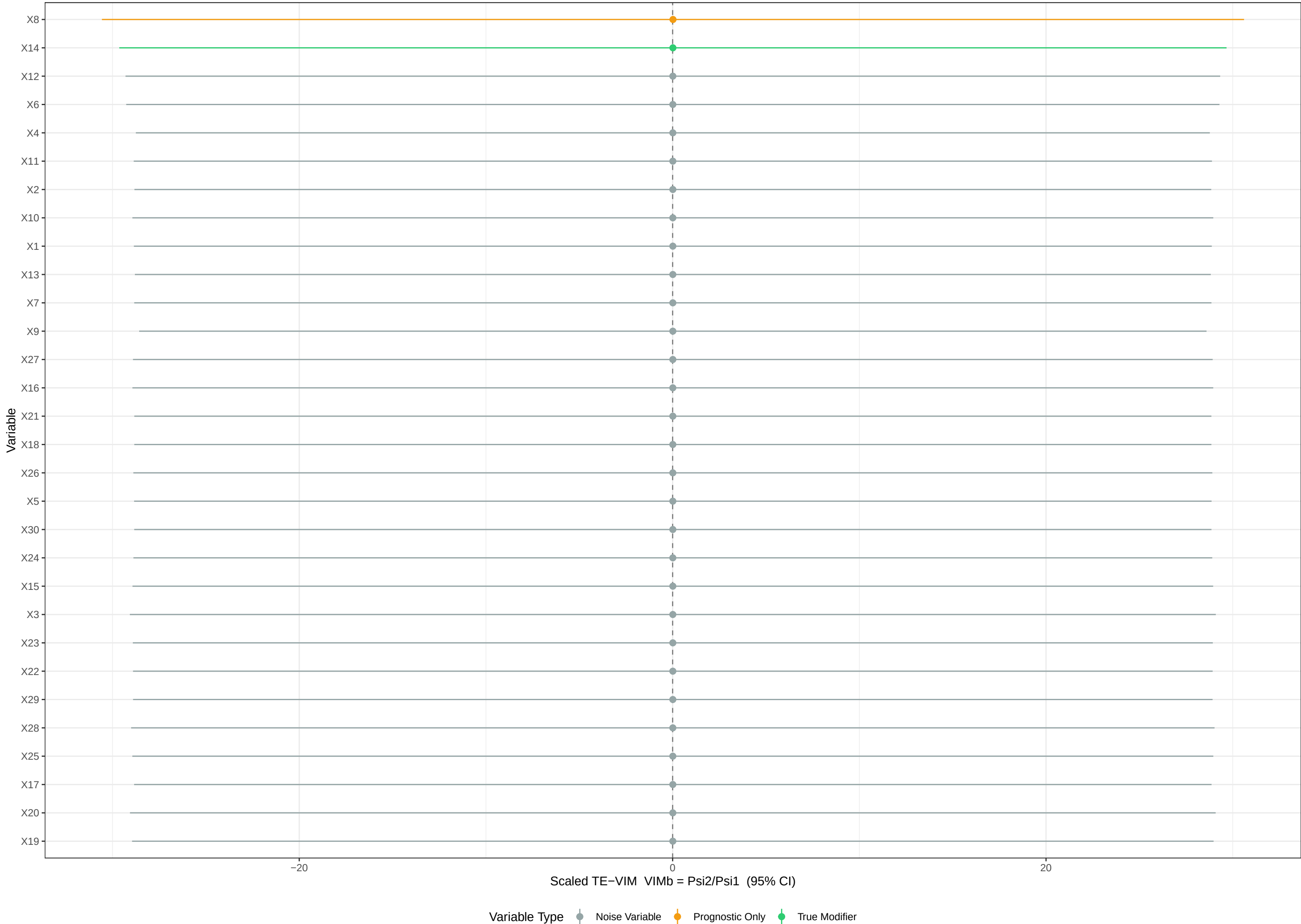
Objective 2: Most Frequently Selected Variables (Scenario 2, $\gamma > 0$)

Top 5 variables selected most often across sims with heterogeneity

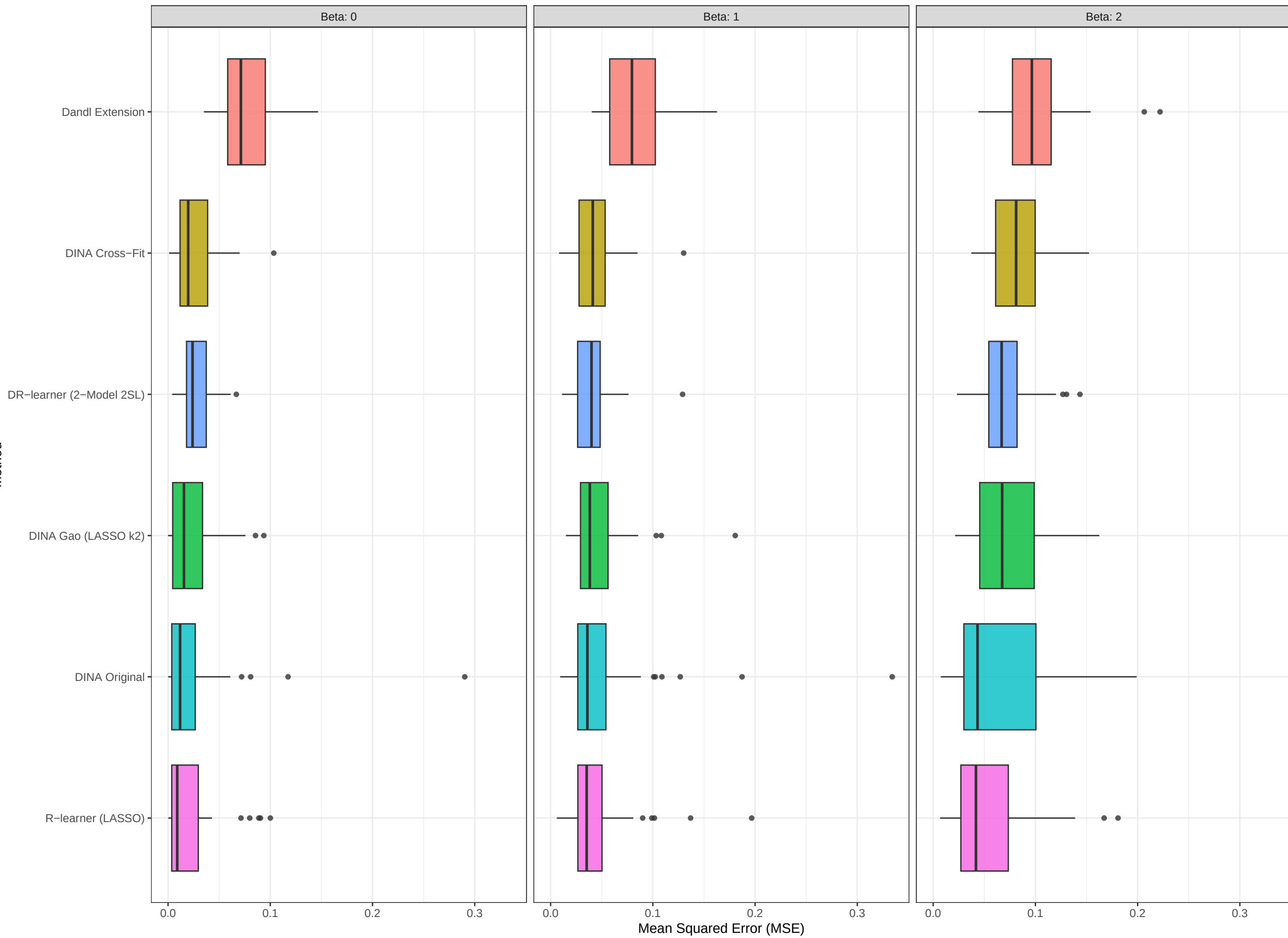


Objective 2 (TE-VIM TMLE): Scaled VIM Ranking (Scenario 2, beta>0)

Li 2026 VIMb = Psi2/Psi1 (TMLE), pooled over sims. True modifier(s): X14

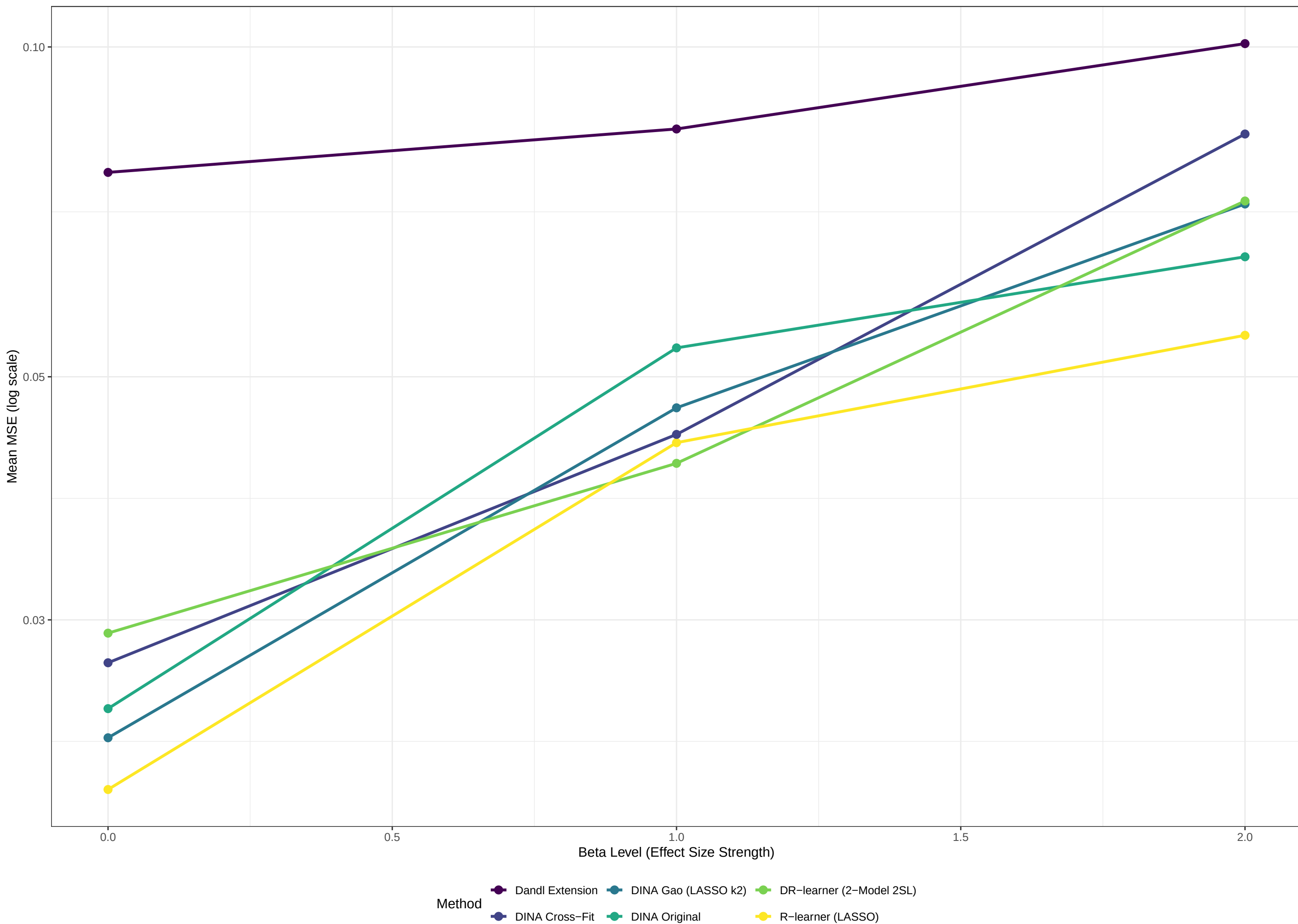


Objective 3: MSE Distribution (Scenario 2)
 Lower MSE = better CATE estimation. Spread = stability.

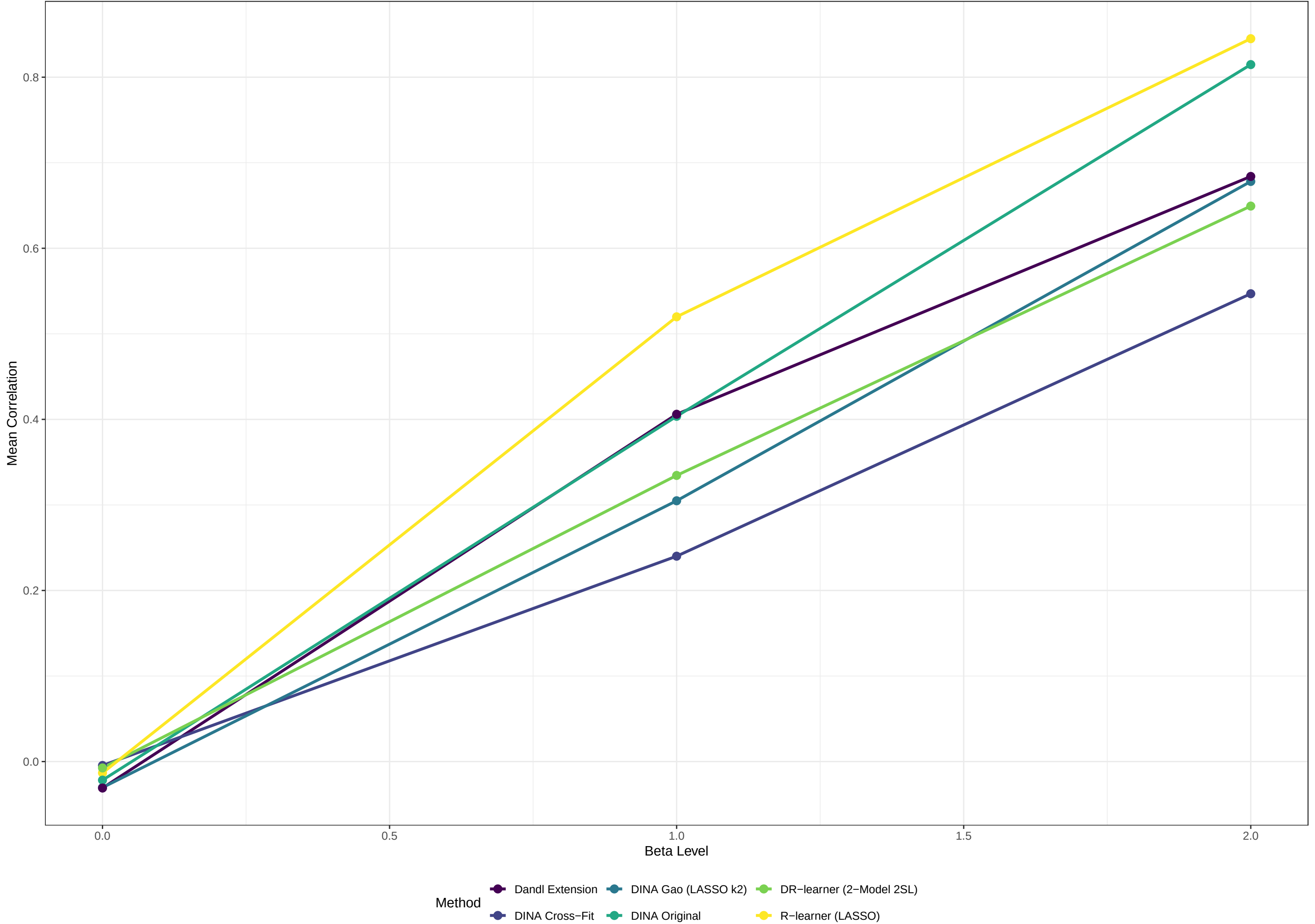


Objective 3: CATE Estimation Performance (Scenario 2)

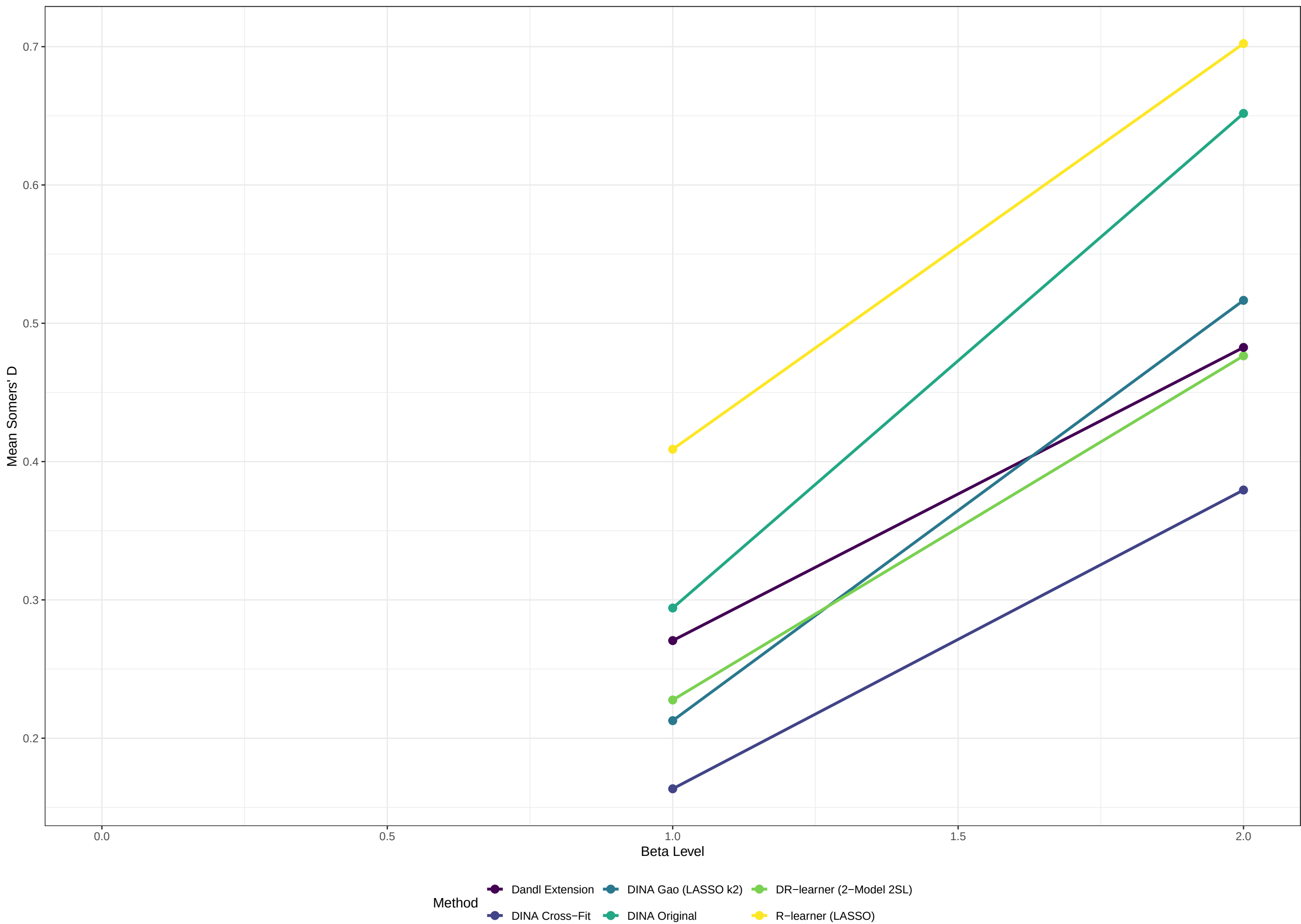
Mean Squared Error for estimating individual treatment effects



Objective 3: Correlation with True Effects (Scenario 2)
Higher correlation = better alignment with true treatment effects



Objective 3: Somers' D Rank Correlation (Scenario 2)
Sechidis et al. (2026) recommended Obj-3 metric; higher = better ranking



Summary Comparison (Scenario 2) – averaged over .>0
True Modifiers: X14

Method	Avg_Power	Avg_Detection	Avg_MSE	Avg_Correlation	Avg_Somers_D	Avg_True_Selection
DINA Gao (LASSO k2)	0.823	50	0.059	0.491	0.365	39
DandI Extension	0.819	50	0.092	0.545	0.377	57
DINA Original	0.819	50	0.059	0.609	0.473	42
DINA Cross-Fit	0.819	50	0.064	0.393	0.271	27
R-learner (LASSO)	0.819	50	0.049	0.682	0.556	34
DR-learner (2-Model 2SL)	0.814	51	0.057	0.492	0.352	40